Design for the Classification of Eavesdropping in Quantum Cryptography

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***Abstract*—Quantum key distributions inherently identify eavesdroppers by virtue of measurement distortion, altering the spin of photons in unpredictable ways. However, the time demands of manual detection strategies make large scale analysis unfeasible. Addressing this, a fully connected neural network was developed to input aggregated data of photon strings, allowing for quick identification of potential security hazards. The model achieves a 91% success rate on simulated BB84 style data, proving that such automation provides fast and scalable identification in quantum communication environments.**

***Index Terms*—quantum key distribution, BB84 protocol, eavesdropping detection, neural networks, photon-level classification**

I. Introduction

The fundamental principle for quantum cryptography lies in the spin behavior of subatomic particles as noted by Griffiths in [1]. Following the percepts of the Copenhagen interpretation of quantum mechanics, a particle's spin component exists as a superposition prior to measurement, collapsing to a definite value only when measured.

Heisenberg's uncertainty principle is given as,

$$\Delta S_x \Delta S_y \geq \frac{\hbar}{2} |\langle S_z \rangle| \tag{1}$$

where ΔS_x and ΔS_y are uncertainties of spin in the x and y directions, $|\langle S_z \rangle|$ is the magnitude of the expectation value of the spin in the z direction, and $\hbar$ is Planck's constant. To illustrate the uncertainty, the case of $\langle S_z \rangle = \pm \frac{\hbar}{2}$ is considered. Substituting this into (1) results in

$$\Delta S_x \Delta S_y \geq \frac{\hbar^2}{4} \tag{2}$$

representing the point of maximum uncertainty. This resultant is some combination of spin in both the x and y directions. This fundamental behavior illustrates the core aspect of quantum cryptography, where measuring one observable in a non-commuting pair unavoidably randomizes the outcome of the other.

For this paper, the data is configured in BB84 style, following the conventions created by Bennett and Brassard in [2], where spins are reduced to rectilinear and diagonal bases, giving a binary differentiation between possible states. When a photon has been previously measured in the rectilinear basis, it collapses down to a spin component along that direction. A subsequent measurement with a diagonal basis would then yield an unknown value of either diagonally up or diagonally down, with both having equal chance. This is displayed in Fig. 1 below.

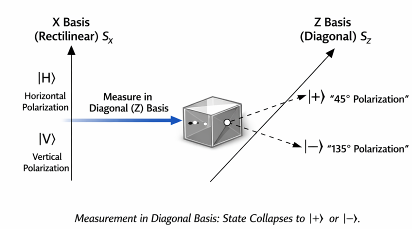

Fig. 1. Measurement in Diagonal Basis: State Collapses to $|+\rangle$ or $|-\rangle$.

Quantum cryptography works by representing data with photon spins, which in turn requires that a recipient of these photons use the correct configuration of measurement filters to interpret it. Otherwise, each filter would alter the data from its original state. However, because of the quantum mechanics involved, if third party eavesdropping was attempted on a transmission, there would be downstream effects on the recipient. The eavesdropper could possibly collapse the spin into a new direction, leading to distortion of the data on the receiving end. By comparing filters and data, the transmitter and receiver could determine that an eavesdropping attempt was made.

Although the BB84 protocol provides a theoretically robust mechanism for detecting eavesdropping, its practical implementation becomes increasingly challenging as photon transmission rates scale into the thousands or millions per second. Detecting intrusion requires analyzing basis agreements, bit-error patterns, and disturbance statistics across large datasets, making manual inspection unfeasible.

II. Background

Supervised learning is a natural choice for this classification problem, as the available features can be mapped to labeled outcomes through a learned functional relationship. In this work, the model used to perform this mapping is a fully connected neural network, which approximates the underlying function by combining input features through weighted linear operations followed by nonlinear activation functions. More specifically, each neuron of the neural network receives the entire input vector and forms a weighted combination of those features. The pre-activation value for a single neuron is

$$z = w^T x + b \tag{3}$$

where w^T is the transpose of the weight vector, x is the input vector, and b is a bias term. This value is then passed through an activation function, which transforms the model from linear to nonlinear. It is important to note that each neuron will connect with all neurons of the preceding layer. An example of the neural network architecture used in this case is shown in Fig. 2.

Through a process termed forward propagation, input data is transformed by passing through each layer of the neural network until reaching an output state in the final node. In binary classification, the output of this will yield a probability after performing a sigmoid activation. Learning occurs through backpropagation, where the network evaluates its prediction using a loss function and computes gradients with respect to each weight. An optimizer such as Adam then updates the weights in the direction that reduces the loss, gradually improving the model's performance over successive training iterations.

To improve performance in generalized cases, a section of validation data is withheld and evaluated against, providing a figure to protect against overfitting. The optimal solution will be a point where the loss of the validation data is at a minimum.

III. Methodology

To implement the eavesdropping detector, a simulated dataset was generated to emulate the spin behavior of photon measurements in the BB84 protocol. The dataset consisted of 500 samples, following the recommendations of Golestaneh *et al* [3]; each sample represents a string of four photons and containing 25 numerical features. These features included the transmitted bit and basis, the received bit and basis, a basis-match indicator, and a bit-error flag for each photon. Each sample was also assigned a binary label indicating whether the sample had been subjected to eavesdropping.

The code was developed in Python 3.13 using NumPy, Pandas, PyTorch, scikit-learn, and Matplotlib. NumPy and PyTorch provided the numerical and tensor-based operations required for constructing and training the neural network. Pandas was used to organize the simulated data into dataframes, which facilitated inspection and verification of correct feature generation. Matplotlib was employed to visualize the training and validation loss curves, allowing for a determination of optimal training episodes. Scikit-learn was used to perform the train–test-validation split of the dataset.

Prior to training, labels were sectioned off of each sample, resulting in a 24 dimensional input vector. This vector was passed into a fully connected neural network consisting of three cascading hidden layers with 16, 12, and 8 neurons, respectively. Each layer used ReLu as the activation function. This outputs to a single neuron with a sigmoid activation for classification of binary probabilities. A graphical representation of this architecture is shown in Fig. 2.

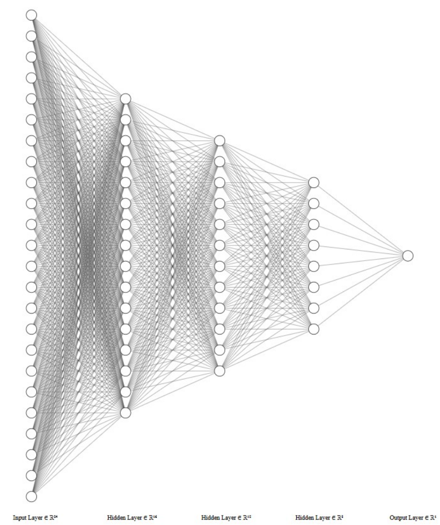

Fig. 2. Neural Network Layer Configuration

Training was performed using binary cross entropy as the loss function for both the training and validation sets. The Adam optimizer was employed to update the network weights, using a learning rate of 0.001 and a batch size of 16. The model was trained for 25 epochs, with a dropout rate of 20% applied during training to reduce overfitting.

IV. Results

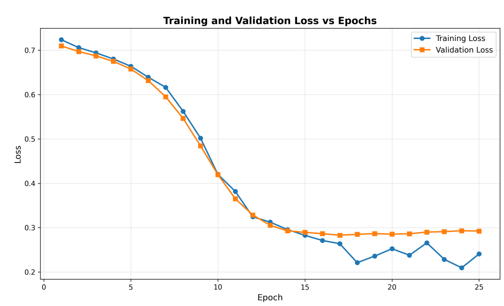

Fig. 3. Train vs. Test Loss Graph

TABLE I
Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.88	1.00	0.94	68
1	1.00	0.72	0.84	32
Accuracy		91%		

As shown in Table I, the model achieved an overall classification accuracy of 91%. From Tables I and II it is evident that

TABLE II
CONFUSION MATRIX

	Predicted 0	**Predicted 1**
Actual 0	68	0
Actual 1	9	23

there was near perfect detection of non-eavesdropped samples and moderate recall on eavesdropped samples.

V. DISCUSSION

As shown in Fig. 3, the training and validation loss curves exhibit smooth and stable convergence, indicating that the selected learning rate and batch size were appropriate. Based on this behavior, 25 epochs was chosen as the optimal training duration, as additional epochs produced a plateau in validation loss and signs of overfitting. The classification results in Tables I and II demonstrate that the model identifies non-eavesdropped samples with high accuracy; however, its performance on eavesdropped samples is limited, reflected by the reduced recall for class 1.

From Table I, there were only 32 total events of eavesdropping that occurred in the test set, raising the possibility that this set was not large enough to draw statistical significance. Additionally, over the course of the generated data set there were only 149 examples of eavesdropped data compared to 351 examples of non-eavesdropped. While this would ideally provide a 50/50 split, the stochastic nature of the data generation did not make this feasible. This disparity may be the culprit behind the higher classification rate for non-eavesdropped samples.

To rectify this, an attempt was made to apply a weight bias to the classification rate to incentivize greater risk taking from the model in flagging eavesdropping, but the resulting model created diverging curves with increasing loss. Additionally, there was an attempt made to remove the basis match and filter error features from the input vector to introduce greater difficulty for the model, but this similarly produced diverging results.

For future iterations of this design, experimentation of larger datasets would prove insightful, allowing for perhaps better training over eavesdropped samples. Expanding the length of the photon strings would also be interesting, potentially allowing for the identification of stress points of the current hardware in terms of data handling capabilities.

REFERENCES

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